

Modelo de Datos - Sistema de Gestión de Vuelos

Convenciones:

Las tablas y columnas usan **snake_case** en **singular**.

Las claves primarias son **UUID** generadas con `gen_random_uuid()`.

Todas las entidades incluyen los siguientes **campos de auditoría**: **status**, **created_at**, **created_by**, **updated_at**, **updated_by**, **deleted_at**, **deleted_by**.

Módulo: Geolocalización

- **continent**: id, code, name, description
- **country**: id, code, name, description, continent_id
- **state**: id, code, name, description, country_id
- **city**: id, code, name, description, state_id

Módulo: Infraestructura

- **airport**: id, code, name, address, city_id
- **terminal**: id, code, name, airport_id
- **boarding_gate**: id, code, name, terminal_id

Módulo: Parametrización

- **document_type**: id, code, name, description
- **flight_type**: id, code, name, description
- **aircraft_type**: id, code, name, description
- **crew_role**: id, code, name, description
- **ticket_class**: id, code, name, description

Módulo: Seguridad

- **role**: id, name, description
- **module**: id, name, color, icon, path
- **form**: id, name, color, icon, path, section
- **form_module**: id, form_id, module_id
- **role_form**: id, role_id, form_id
- **person**: id, first_name, last_name, document_number, birth_date, gender, phone, email, address, attendant_name, attending_phone, document_type_id, city_id
- **user_account**: id, username, password_hash, person_id, role_id

Módulo: Talento Humano

- **employee**: id, salary, hire_date, crew_role_id, person_id

Módulo: Gestión de Aeronaves

- **aircraft**: `id`, `manufacturer`, `registration_code`, `hours_in_use`, `aircraft_type_id`

Módulo: Operaciones de Vuelo

- **flight**: `id`, `flight_date`, `departure_time`, `arrival_time`, `flight_type_id`, `origin_boarding_gate_id`, `destination_boarding_gate_id`, `aircraft_id`
- **crew_assignment**: `id`, `flight_id`, `employee_id`, `crew_role_id`
- **ticket**: `id`, `code`, `flight_id`, `passenger_id`, `ticket_class_id`, `seat_number`, `price`
- **baggage**: `id`, `code`, `weight`, `ticket_id`

Módulo: Pasajeros y Servicios

- **passenger**: `id`, `frequent_flyer_number`, `person_id`

Módulo: Notificaciones

- **message_template**: `id`, `code`, `name`, `subject_template`, `body_template`
- **notification**: `id`, `person_id`, `channel`, `subject`, `message`, `sent_at`, `status`

Flight Management System – Refined Data Model (v2)

🕒 Conventions

- Tables and columns use **snake_case**, in **singular**.
 - Primary keys are **UUID** generated with `gen_random_uuid()` (`CREATE EXTENSION pgcrypto;`).
 - **Audit fields (7)**:
 - `status`
 - `created_at` / `created_by`
 - `updated_at` / `updated_by`
 - `deleted_at` / `deleted_by`
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Module: Geolocation

- **continent** {`id`, `code`, `name`, `description`}
- **country** {`id`, `code`, `name`, `description`, `continent_id`}
- **state** {`id`, `code`, `name`, `description`, `country_id`}
- **city** {`id`, `code`, `name`, `description`, `state_id`}

Module: Infrastructure

- **airport** {`id`, `code`, `name`, `address`, `city_id`}
- **terminal** {`id`, `code`, `name`, `airport_id`}
- **boarding_gate** {`id`, `code`, `name`, `terminal_id`}

Module: Parameterization

- **document_type** {`id`, `code`, `name`, `description`}
- **flight_type** {`id`, `code`, `name`, `description`}

- **aircraft_type** {id, code, name, description}
- **crew_role** {id, code, name, description}
- **ticket_class** {id, code, name, description}

Module: Security

- **role** {id, name, description}
- **module** {id, name, color, icon, path}
- **form** {id, name, color, icon, path, section}
- **form_module** {id, form_id, module_id}
- **role_form** {id, role_id, form_id}
- **person** {id, first_name, last_name, document_number, birth_date, gender, phone, email, address, attendant_name, attending_phone, document_type_id, city_id}
- **user_account** {id, username, password_hash, person_id, role_id}

Module: Human Resources

- **employee** {id, salary, hire_date, crew_role_id, person_id}

Module: Aircraft Management

- **aircraft** {id, manufacturer, registration_code, hours_in_use, aircraft_type_id}

Module: Flight Operations

- **flight** {id, flight_date, departure_time, arrival_time, flight_type_id, origin_boarding_gate_id, destination_boarding_gate_id, aircraft_id}
- **crew_assignment** {id, flight_id, employee_id, crew_role_id}
- **ticket** {id, code, flight_id, passenger_id, ticket_class_id, seat_number, price}
- **baggage** {id, code, weight, ticket_id}

Module: Passengers & Services

- **passenger** {id, frequent_flyer_number, person_id}

Module: Notifications

- **message_template** {id, code, name, subject_template, body_template}
- **notification** {id, person_id, channel, subject, message, sent_at, status}